

[EVENT]

# HANDS-ON RAG

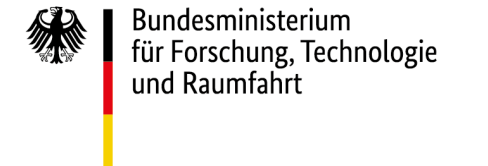
Build your own document Q&A system

---

Felix Boelter | [Co-moderator]

AI Service Centre Berlin-Brandenburg | [Date]

Gefördert durch:



Design IT.  
Create Knowledge.

[www.hpi.de](http://www.hpi.de)



# PARTICIPANT LIST

The list is going round now. Please sign in while everyone is still seated.

| Name | Institution | Email |
|------|-------------|-------|
|      |             |       |
|      |             |       |
|      |             |       |

The printout stays with the workshop lead. Please hand it back at the end.

## START THE INSTALL NOW

It downloads about 2 GB, so let it run while we talk.

```
git clone https://github.com/  
    aihpi/workshop-rag-basics.git  
cd workshop-rag-basics/workshop  
uv sync
```



Needs uv and Git, from the  
workshop-getting-started guide.

01

# AI SERVICE CENTRE BERLIN-BRANDENBURG

---

Who we are and what you can use.

Gefördert durch:

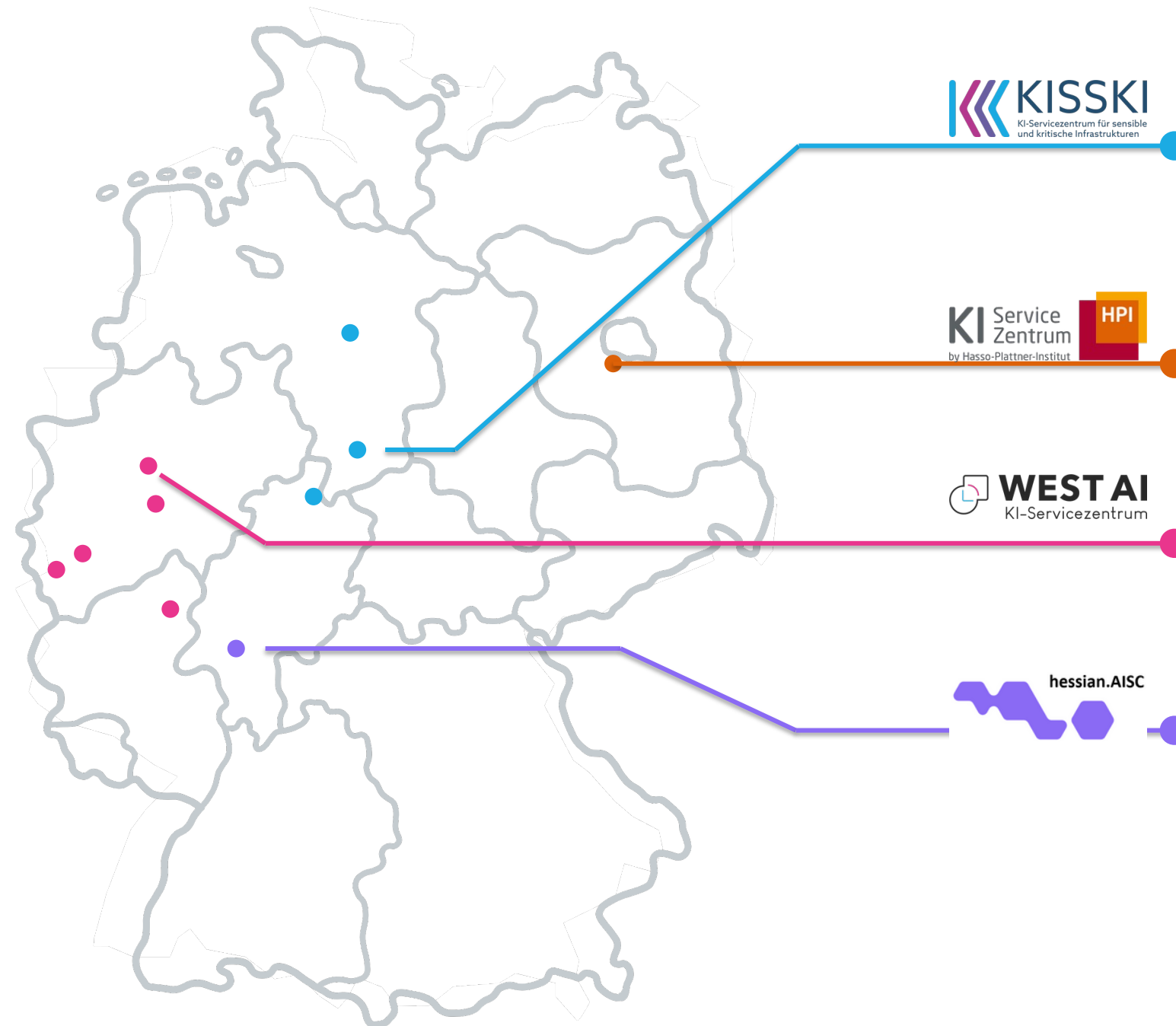


Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

# FOUR AI SERVICE CENTRES



[hpi.de/ki-servicezentrum](https://hpi.de/ki-servicezentrum)



**KISSKI**  
KI-Servicezentrum für sensible  
und kritische Infrastrukturen

**Göttingen | Kassel | Hannover**

Sensitive and critical infrastructure, medicine & energy

**KI Service Zentrum**  
by Hasso-Plattner-Institut

**Potsdam | Berlin**

Education and consulting, AI in business & society

**WEST AI**  
KI-Servicezentrum

**Bonn | St. Augustin | Aachen | Jülich | Dortmund**

AI hardware, consulting and training; multimodal and transferable models

**hessian.AISC**

**Darmstadt**

Explainability, generalisability and contextual adaptation



**KI-Servicezentren**

Gefördert durch:



Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

**KI Service Zentrum**  
by Hasso-Plattner-Institut



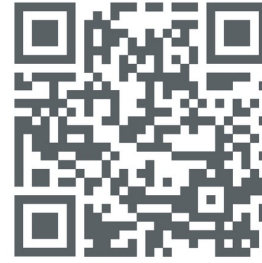
**Goal: lower the barriers to putting AI to work in society and business**

# EDUCATION



Newsletter

## Talks



[tele-task.de/series/1463](https://tele-task.de/series/1463)

- Guest talks on research and innovation

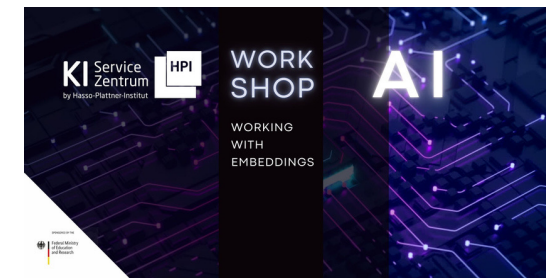


## Workshops



[aimaker.community](https://aimaker.community)

- Hands-on topics
- For example: speech2summary, Docker for ML, semantic search

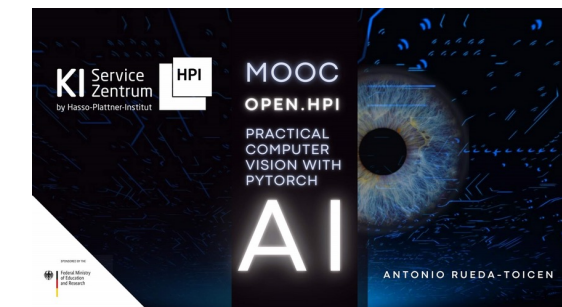


## MOOCs



[open.hpi.de/channels/ai-service-center](https://open.hpi.de/channels/ai-service-center)

- ChatGPT: what does generative AI mean for society?
- Profitable AI
- Understanding and avoiding AI bias



# INFRASTRUCTURE



[aisc.hpi.de](https://aisc.hpi.de) · request access



- Access **free of charge**
- No production use
  - Data should be **anonymised or synthetic**
  - No **hosting of products**
- **Reporting & publication** by users
- **Existing rights** stay with users
- **New rights** stay with users
  - Usage rights granted for research and teaching

## Training

- 64 NVIDIA H100 GPUs

## Inference

- 40 NVIDIA A30 GPUs

## ARM server

- Ampere Altra Max M128-30 CPU
- 2× NVIDIA L40 GPUs

## GPU server

- AMD Epyc CPU
- 8× NVIDIA L40S GPUs

## Edge

- ARMv8 CPU
- NVIDIA Jetson AGX modules

## Neuromorphic

- 288 SpiNNaker2 chips

## Storage

- 1.5 PB NVRAM

## Network

- 400 Gb/s InfiniBand
- 200 Gb/s Ethernet

# CONSULTING



Book office hours

## AI office hours

- AI infrastructure
- AI models & frameworks
- AI use cases

## AI pilot projects

- Co-developing a prototype
- Applications every three months
- Selection criteria such as AI maturity and public benefit
- Results are published

## Collaborations

- Jointly organised network meetings

## Past AI pilot projects

- Generating maths problems
- Generating plain language
- Generating upcycling ideas
- Reducing food waste
- Dating documents by handwriting



Apply now!

# BY 1:30 YOU HAVE

Gefördert durch:



Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

KI Service  
Zentrum  
by Hasso-Plattner-Institut



- 1 A chatbot on your laptop that answers from your own PDF.
- 2 Every claim links to the page it came from, with a score that says how well it is backed.
- 3 A list of what to check before you trust it.

## AGENDA

| 0:00                   | 0:17     | 0:30          | 0:43       | 0:56            | 1:12                    |
|------------------------|----------|---------------|------------|-----------------|-------------------------|
| Welcome · AISC · setup | 1 · Chat | 2 · Documents | 3 · Search | 4 · RAG + score | 5 · Your PDF · transfer |

Every notebook block: a minute of slides, then the notebook, then a quick check.

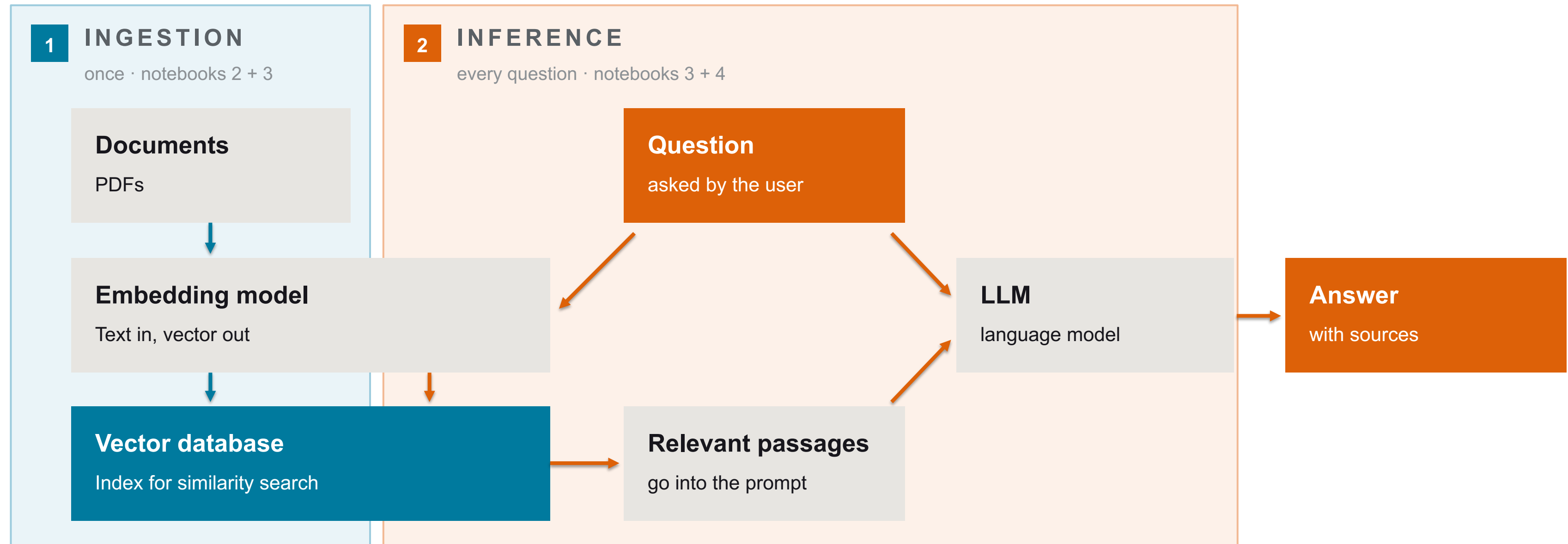
# ARCHITECTURE

Gefördert durch:



Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

KI Service  
Zentrum  
by Hasso-Plattner-Institut



**Once, ahead of time,** the documents are read and cut into chunks. The embedding model turns every chunk into a vector, a list of numbers that captures its meaning. The vectors go into the vector database.

**For every question,** the same model turns the question into a vector and searches the same database. The closest chunks go to the language model with the question, and it answers with sources.

# SETUP

## 1 Install (started on slide 2)

```
git clone https://github.com/aihpi/workshop-rag-basics.git
cd workshop-rag-basics/workshop
uv sync
```

## 2 Add the key

```
cp .env.example .env # then paste the key from the whiteboard
```

## 3 Check, and fetch Docling's models

```
uv run python check_setup.py
```

## 4 Start

```
uv run jupyter lab
```

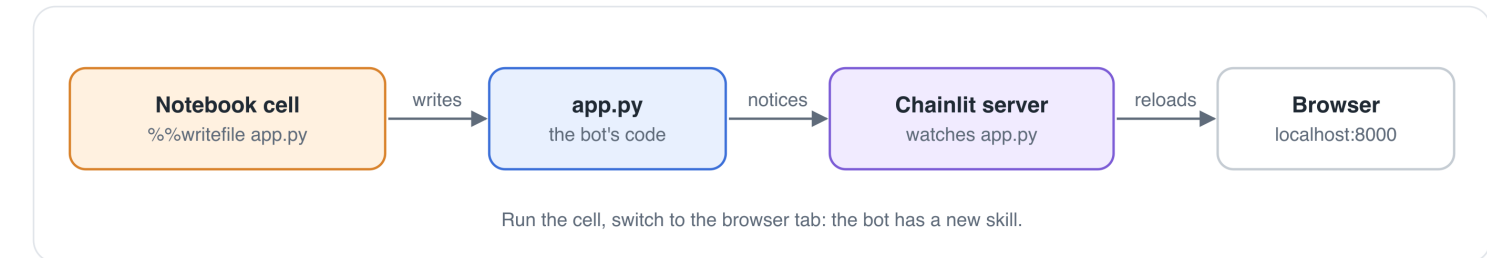
Done when check\_setup.py prints **You're ready for the workshop.**

Gefördert durch:



Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

KI Service  
Zentrum  
by Hasso-Plattner-Institut



Cells that start with `%%writefile app.py` write the chatbot. Chainlit notices the change and reloads it in your browser at `localhost:8000`.

### CHAT MODEL

gemma-4-31b

### EMBEDDING MODEL

octen-embedding-8b

### RUNS ON

our own infrastructure in Potsdam

# TALKING TO AN LLM

## NOTEBOOK 1 · 6 MIN

### IN THIS NOTEBOOK

- Connect to the model through the gateway
- Ask about a detail from one of our papers
- Build a chatbot that remembers and streams

### YOUR TURN · IN THE CHATBOT

- 1 Personality: change the system prompt
- + Bonus: stopwatch, you'll want it in notebook 4

Solutions are folded away under each exercise.

Gefördert durch:



Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

KI Service  
Zentrum  
by Hasso-Plattner-Institut



```
localhost:8000
```

In Kage et al. (2018), what were the lifetime-encoded beads loaded with?

✓ The model answers. Is it right?

# WAS IT RIGHT?

Hands up.  
Remember your letter.

## NOTEBOOK 1

The model answered the Kage question. Was it ...

- A right
- B wrong
- C partly right
- D can't tell

# HALF RIGHT, AND YOU CAN'T TELL WHICH HALF

Gefördert durch:



Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

KI Service  
Zentrum  
by Hasso-Plattner-Institut

HPI

01

## No access to your documents

Language models are trained mostly on public data. Your lab reports and paywalled papers are unknown to them.

02

## Similar instead of right

When the right information is missing, the model falls back on related material: the rules for no-parking zones answer a question about no-stopping zones.

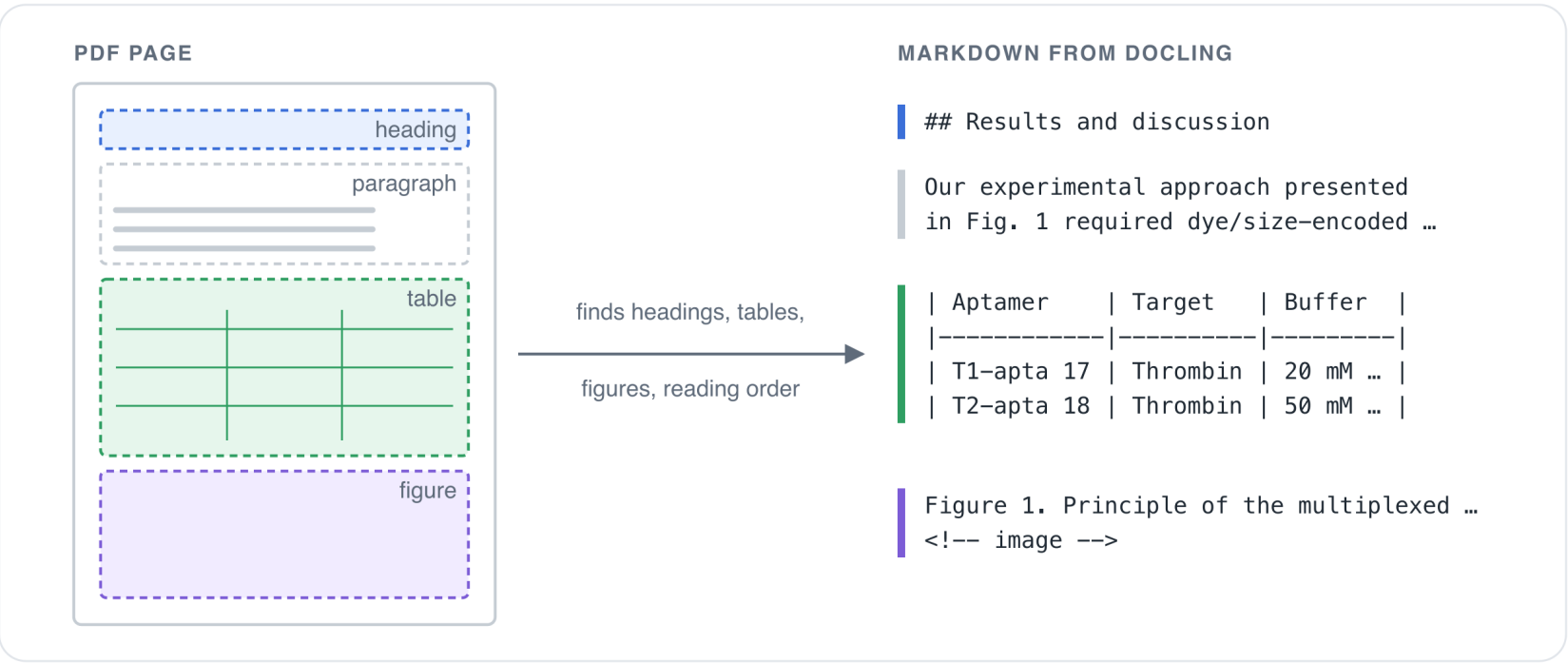
03

## RAG decides what the answer is built on

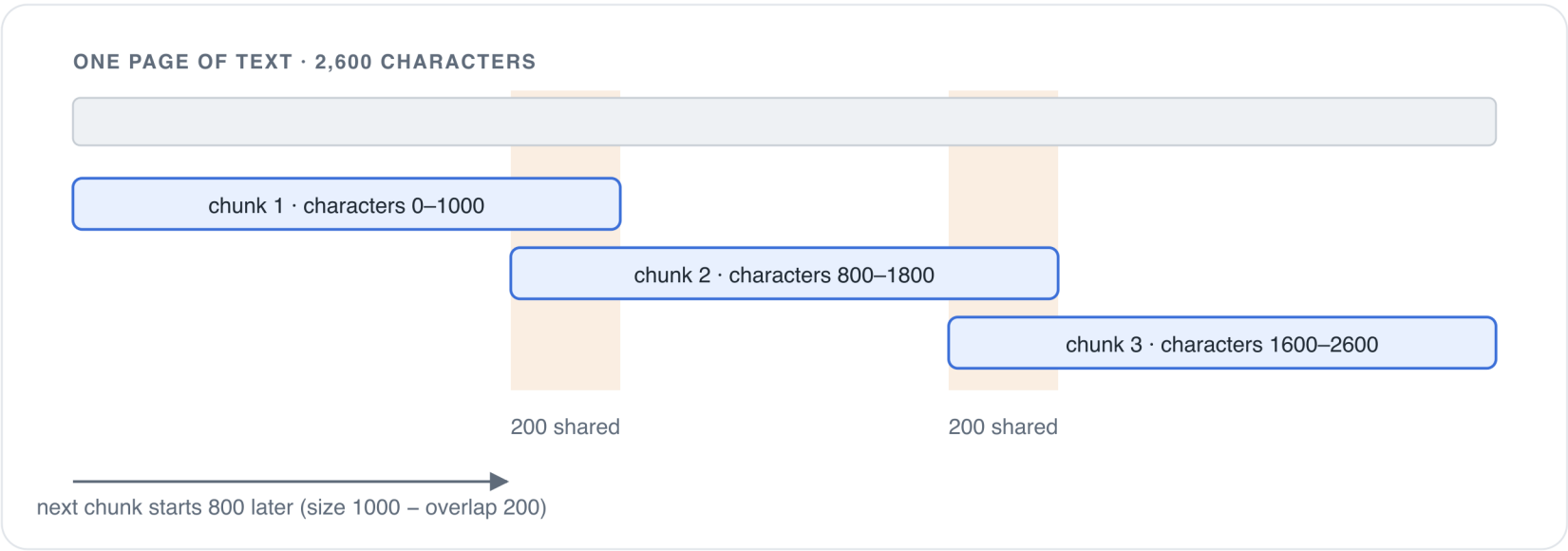
Retrieval-augmented generation sets which documents an answer may use, and cites the page.

# READING DOCUMENTS

NOTEBOOK 2 · 7 MIN



A PDF stores where to draw each letter. Docling finds headings, paragraphs and tables, and writes Markdown.



We cut chunks of 1000 characters; a new one starts every 800, so neighbours share 200.

## YOUR TURN · IN THE CHATBOT

- 1 Where are the cuts? Try 300 and 3000
- + Bonus: only the tables

# WHAT DID YOU FIND?

Vote by show of hands.  
Answers on the next slide.

## NOTEBOOK 2

**What happened to the tables when Docling read the papers?**

- A They were dropped
- B They became loose lines of numbers
- C They stayed tables, in Markdown

## NOTEBOOK 2 · YOUR TURN

**What happens with smaller chunks, 300 characters instead of 1000?**

- A Fewer chunks, and tables stay whole
- B More chunks, and more of them cut through a table
- C Nothing changes

# WHAT WE FOUND

## NOTEBOOK 2

What happened to the tables when Docling read the papers?

**C** They stayed tables, in Markdown

All 5 tables in the three papers came through with their rows and columns, searchable like any other text.

## NOTEBOOK 2 · YOUR TURN

What happens with smaller chunks, 300 characters instead of 1000?

**B** More chunks, and more of them cut through a table

628 instead of 197 chunks, and 17 instead of 5 end in the middle of a table row. At 3000: 69 chunks, none.

NEXT

We have good chunks. How do we find the right ones for a question?

# SEARCH

## NOTEBOOK 3 · 8 MIN

### IN THIS NOTEBOOK

- Embed sentences and compare how similar they are
- Store all chunks in Qdrant, in memory
- Turn the bot into a search engine, no LLM yet

### YOUR TURN · IN THE CHATBOT

**1** Fewer, better results: pick a score cutoff

**+** Bonus: ask in German

Solutions are folded away under each exercise.

Gefördert durch:



Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

KI Service  
Zentrum  
by Hasso-Plattner-Institut

HPI

localhost:8000

What were the lifetime-encoded beads loaded with?

Kage\_2018\_SciReports.pdf, p. 2 (score 0.73)  
› ... loaded with different organic fluorophores ...

Kage\_2018\_SciReports.pdf, p. 3 (score 0.64)  
› Table 1. Lifetime codes and respective luminophores ...

What is the capital of France?

Kage\_2018\_SciReports.pdf, p. 6 (score 0.17) ...

# WHY DID FRANCE GET A HIT?

Gefördert durch:

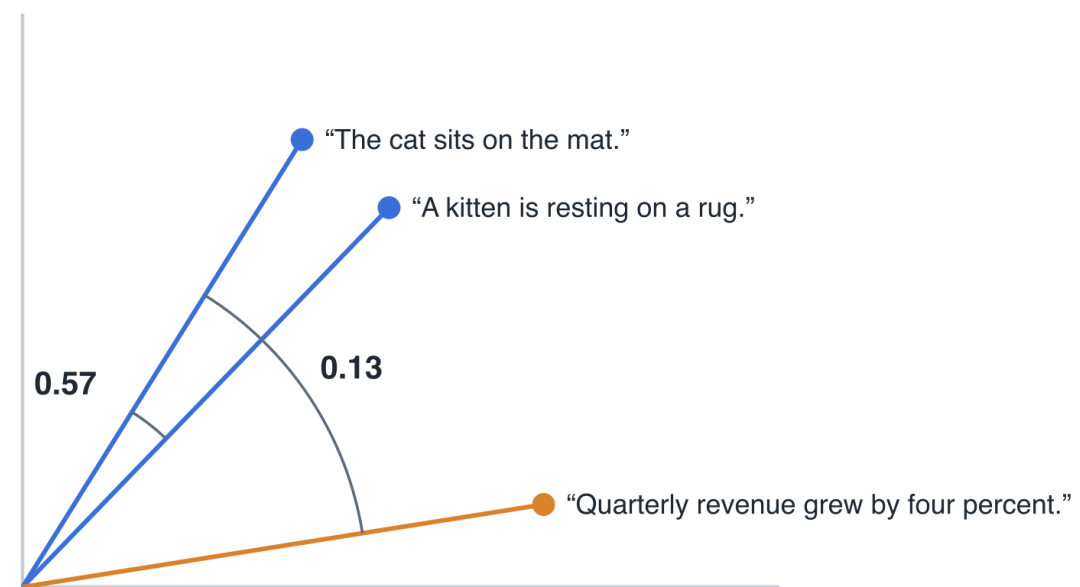


Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

KI Service  
Zentrum  
by Hasso-Plattner-Institut



Similar meaning → arrows point the same way.



Similarity = how closely two arrows point the same way (1 = same direction).  
Real embeddings have 4096 dimensions; this drawing has 2.

## Collection “papers”

| id | VECTOR                      | PAYLOAD                           |
|----|-----------------------------|-----------------------------------|
| 0  | [0.012, -0.031, 0.087, ...] | Kage_2018 · p. 1 · “Time-res...”  |
| 1  | [-0.007, 0.044, 0.019, ...] | Kage_2018 · p. 1 · “multiplex...” |
| 2  | [0.021, 0.003, -0.058, ...] | Kage_2018 · p. 2 · “are loade...” |

... 203 points, one per chunk

query\_points

**Question → vector**  
embedded like the chunks

returns

**5 closest points**  
each with score + payload

vector = what Qdrant searches by

payload = what you get back

- 1 An embedding model turns text into a list of numbers, 4096 with ours. Similar meaning, similar direction.
- 2 Search returns the closest chunks, however far away they are.
- 3 Qdrant stores one point per chunk: the vector, and a payload with text, file and page.

# RETRIEVAL-AUGMENTED GENERATION

NOTEBOOK 4 · 11 MIN

## RETRIEVE

the 5 closest chunks

## AUGMENT

the prompt with them, numbered [1] to [5]

## GENERATE

the answer, citing those numbers

Every [n] leads to a chunk, and every chunk remembers its file and page.

## YOUR TURN

- 1 How much context? Score limit 2 and limit 15
- + Bonus: strict or not
- + Bonus: show the evidence; follow-up questions

Gefördert durch:



Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

KI Service  
Zentrum  
by Hasso-Plattner-Institut

HPI

### WHAT THE LLM RECEIVES

system

Answer only from the context. If it isn't there, say you don't know.

user

Context:

[1] ...microbeads are loaded with different organic...  
[2] ...tailor-made melamine beads with a polyelectr...  
:  
[5] ...imitations of lifetime encoding in flow with...

Question: what were the beads loaded with?

LLM  
gemma-4-31b

The beads were loaded with different organic fluorophores with lifetimes in the nanosecond range [1].

green: the 5 chunks Qdrant found · blue: what you typed

localhost:8000

What were the beads in Kage et al. loaded with?

PMMA beads stained with organic dyes from PolyAn GmbH [1], and melamine beads loaded with CdSe/CdS/ZnS quantum dots [1], [2].

Sources: Kage\_2018\_SciReports, p. 2

# WHICH HALF?

THE NOTEBOOK 1 ANSWER, JUDGED IN NOTEBOOK 4 (ONE RUN)

- ✓ The beads were loaded with quantum dots (QDs).
- ✗ Three different types of quantum dots, to create distinct lifetime signatures.
- ✗ One type had a short lifetime of about 10 ns.
- ✗ One type had a medium lifetime of about 25 ns.
- ✗ One type had a long lifetime of about 50 ns.
- ✗ The beads were used as fiducial markers to correct for drift during imaging.

faithfulness **0.17**

**Which part of the system knows the page number of a chunk?**

- A the LLM
- B the payload in Qdrant
- C the embedding model

# PUTTING IT TOGETHER

NOTEBOOK 5 · 6 MIN

Gefördert durch:



Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

KI Service  
Zentrum  
by Hasso-Plattner-Institut

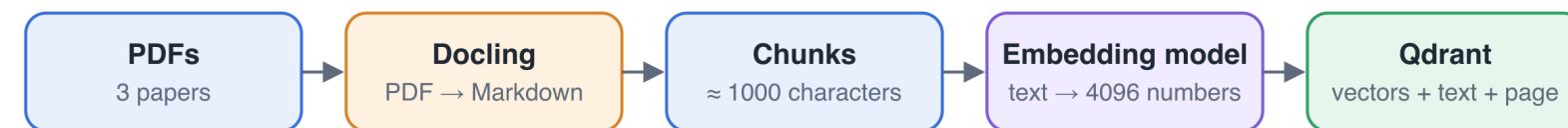
HPI

## YOUR TURN · IN THE CHATBOT

**1** Your own document: attach a PDF, ask, check the page and the score

**+** Bonus: chat with this document only

ONCE, AHEAD OF TIME



searches the same collection

FOR EVERY QUESTION



□ data   □ processing step   □ AI model   □ vector database

## PIECE · NOTEBOOK

Read a PDF with Docling

2

Cut into chunks

2

Turn text into vectors

3

Store and search in Qdrant

3

Answer from the chunks

4

Cite the page

4

Score the answer

4

Your own PDF

5

**Your PDF goes to the AISC gateway in Potsdam. Test or public documents only.**

# YOUR COLLECTION

Gefördert durch:



Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

KI Service  
Zentrum  
by Hasso-Plattner-Institut



## QUESTION 1

**Which collection would you point this at?**

## QUESTION 2

**What do you expect to break first, and why?**

One sentence each.

OFTEN

Scanned PDFs · tables · questions across several documents

# WHERE TO TAKE THIS

Gefördert durch:



Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

**KI** Service  
Zentrum  
by Hasso-Plattner-Institut



## Today

Fixed 1000-character chunks

Qdrant in memory, rebuilt on every start

Vector search only

Faithfulness, one answer at a time

## The template

Structure-aware chunkers, chosen per data source

Qdrant server, ingested once, updated when files change

Hybrid search: vectors plus keyword matching

An evaluation app that compares setups



### The template

[aihpi/pilotproject-rag-template](#) on  
GitHub



### Office hours

Stuck on your own collection? Book a  
slot.



### Pilot projects

Build it with us; applications every three  
months.

Gefördert durch:



# YOUR FEEDBACK COUNTS

We shape our next workshops around your answers. Two minutes is enough.



[kisz@hpi.de](mailto:kisz@hpi.de)

[hpi.de/kisz](https://hpi.de/kisz)

Felix Boelter | [Co-moderator]

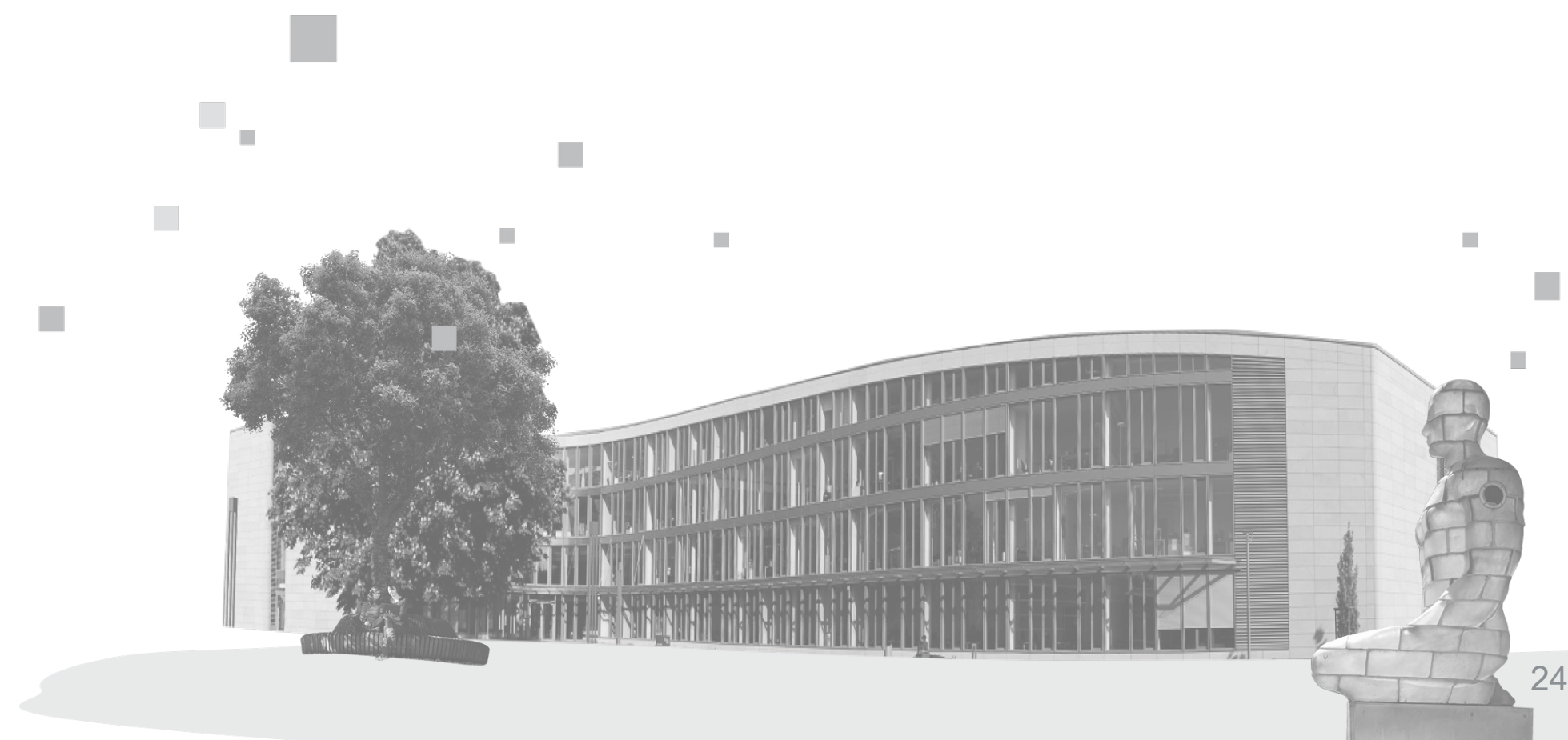
## KEEP GOING

[github.com/aihpi/workshop-rag-basics](https://github.com/aihpi/workshop-rag-basics) · this workshop

[github.com/aihpi/pilotproject-rag-template](https://github.com/aihpi/pilotproject-rag-template) · template

**Design IT.  
Create Knowledge.**

[www.hpi.de](https://www.hpi.de)



# APPENDIX

---

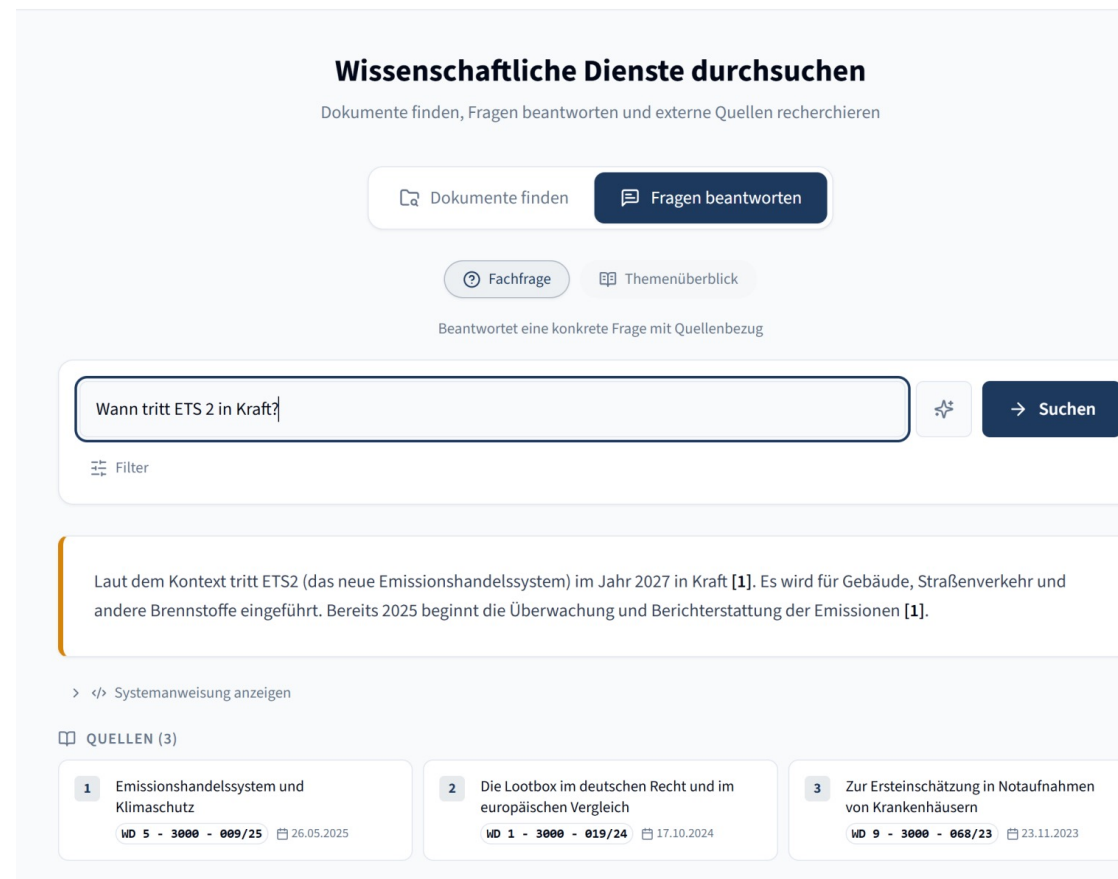
Background for questions. Not part of the 90 minutes.

Gefördert durch:



Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

# RAG IN PRACTICE



# SENTRA

Research Services of the German Bundestag: semantic search across expert reports, an AI summary, every statement traced to its report.

[github.com/aihpi/pilotproject-sentra](https://github.com/aihpi/pilotproject-sentra)

**Semantic search · an AI summary · sources for transparency.** Today you build the same thing, small.

Gefördert durch:



**Bundesministerium  
für Forschung, Technologie  
und Raumfahrt**

**KI Service Zentrum**  
by Hasso-Plattner-Institut



Was ist der Unterschied zwischen Prozess- und Systembausteinen?

Used rag\_retrieve

❖ Prozess-bausteine beschreiben **Methoden, Abläufe und organisatorische Maßnahmen** des Informationssicherheits-Managements. Sie sind nach den Schichten ISMS, ORP, CON, OPS und DER gegliedert und geben jeweils Gefährdungs- sowie Anforderungsübersichten für die jeweiligen Prozesse (z. B. Risikomanagement, Notfall- und Incident-Response) (

QUELLE 1: IT-GRUNDSCHUTZ-BAUSTEINE (S.21) ).

System-bausteine dagegen fokussieren auf **technische Komponenten und deren Umgebung**. Sie ordnen sich den Schichten INF, NET, SYS, APP und IND zu und enthalten Beschreibungen von Infrastruktur-, Netzwerk-, System-, Anwendungs- und Industrie-IT Komponenten samt zugehöriger Gefährdungen und konkreter Sicherheitsanforderungen ( QUELLE 2: DIE IT-GRUNDSCHUTZ-BAUSTEINE (S.132-133) ).

Beide Bausteintypen sind modular aufgebaut, ermöglichen eine gezielte **Bausteinauswahl** und bilden zusammen das Schichten-Modell des IT-Grundschutz-Kompodiums (

QUELLE 3: BAUSTEINE (S.35) ).

**Bezug zu Ihren Interessen:** Der Unterschied ist zentral für Ihre Arbeit an Prozess- und Systembausteinen sowie bei der Auswahl passender Bausteine im Informationsverbund.

Quellen anzeigen

Welche Kriterien nutzen Sie üblicherweise bei der Auswahl von Prozess-bausteinen für Ihr ISMS?

Wie integrieren Sie System-bausteine in die G↓architektur Ihres Informationsverbundes?

Type your message here..

Nutze `/export all` um alle Chats zu exportieren.



LLMs can make mistakes. Check important info



# GrundschutzKI

IT baseline protection: questions in plain language, a short answer with a jump link, the BSI original open next to it.

[github.com/aihpi/pilotprojekt-GrundschutzKI](https://github.com/aihpi/pilotprojekt-GrundschutzKI)

# FROM DOCUMENT TO CHUNKS

Gefördert durch:

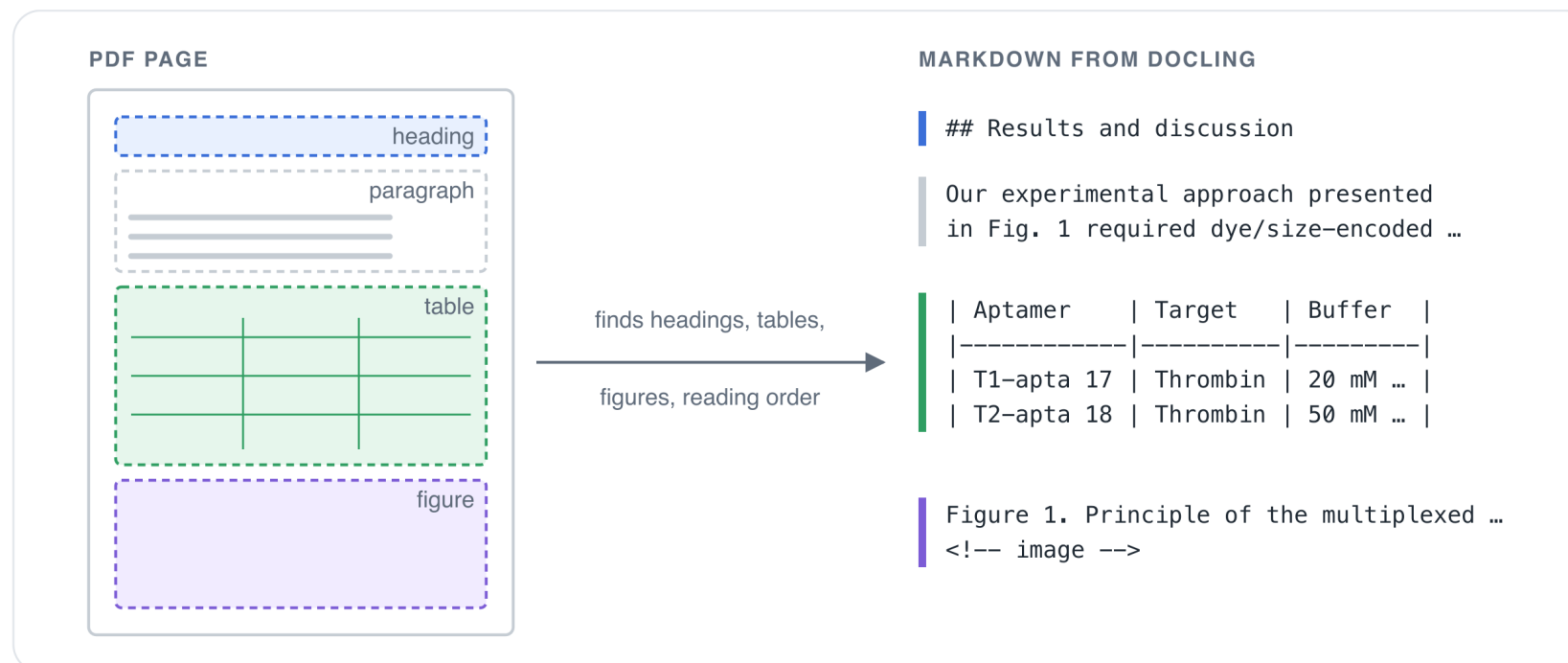


Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

KI Service  
Zentrum  
by Hasso-Plattner-Institut

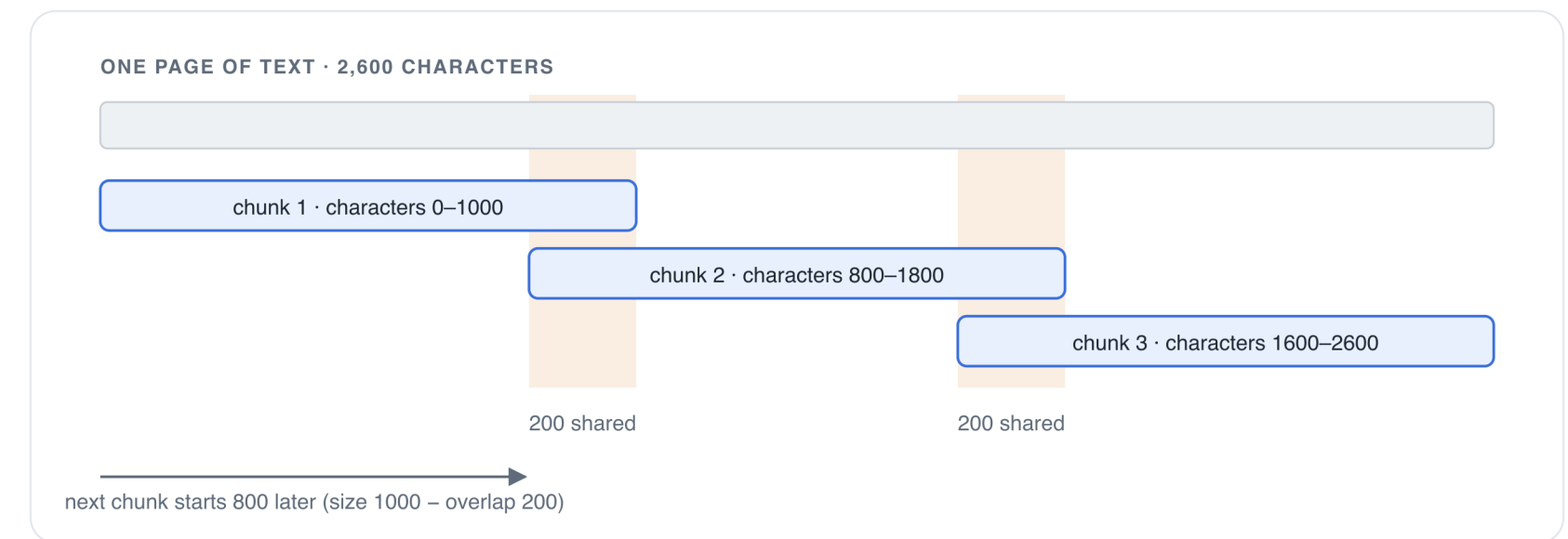


## STEP 1 · DOCLING READS THE PAGE



A PDF only stores where to draw each letter. Docling finds headings, paragraphs, tables and figures, and writes Markdown.

## STEP 2 · CUT INTO CHUNKS



**Our notebook:** every 1000 characters, with 200 overlap. Simple, the same for every document, but it may cut through a table.

**GrundschutzKI:** at every heading, because the BSI standards are clearly structured. A chunk keeps its title. Docling ships several such chunkers.

# PROMPT AND CITATIONS

Gefördert durch:

Bundesministerium  
für Forschung, Technologie  
und RaumfahrtKI Service  
Zentrum  
by Hasso-Plattner-Institut

## RETRIEVE

the 5 closest chunks

## AUGMENT

the prompt with them, numbered [1] to [5]

## GENERATE

the answer, citing those numbers

Every [n] leads back to a chunk, and every chunk remembers its file and page. That is all a citation is.

### WHAT THE LLM RECEIVES

#### system

Answer only from the context. If it isn't there, say you don't know.

#### user

Context:

[1] ...microbeads are loaded with different organic...

[2] ...tailor-made melamine beads with a polyelectr...

⋮

[5] ...imitations of lifetime encoding in flow with...

Question: what were the beads loaded with?

#### LLM

gemma-4-31b

The beads were loaded  
with different organic  
fluorophores with  
lifetimes in the  
nanosecond range [1].

green: the 5 chunks Qdrant found · blue: what you typed

...loaded with different  
organic fluorophores  
in the nanosecond  
range [1]

finds [1]

hits[0].payload  
source: Kage\_2018\_...pdf  
page: 2

cl.Pdf

opens at page 2

The number in the answer points into the list of chunks, and every chunk remembers its file and page.

# IS THE ANSWER ANY GOOD?

Gefördert durch:



Bundesministerium  
für Forschung, Technologie  
und Raumfahrt

KI Service  
Zentrum  
by Hasso-Plattner-Institut



You scored faithfulness in notebook 4. Changed the chunk size or the number of chunks? Measure whether it helped. Neither score needs a hand-written correct answer.

## Faithfulness

Is every claim backed by the retrieved chunks?

```
supported claims ÷ all claims
```

A judge model splits the answer into claims and checks each one against the chunks. 0.5 means half the claims have no source.

## Relevance

Does the answer address the question?

```
similarity(questions from the answer, real question)
```

A judge model writes the questions this answer would fit; their embeddings are compared with the question that was asked.

**In the template:** both scores for every answer, a badge in the chat, and an evaluation app that compares configurations. Built on Ragas.